

Part I. Random and Stochastic ODE

Sec 1. Introduction.

• Einstein 1905.

• Langevin, $\frac{dx}{dt} = f(x) + \underbrace{\sigma(x) \eta_t}$.

$$E(\eta_t) = 0 \quad \underbrace{E(\eta_{t_1} \eta_{t_2}) = \rho \delta(t_1 - t_2)}$$

• Itô: stochastic ODE.

Sec 1. - 1) simple numeric scheme

2) Taylor expansion. — $\left. \begin{array}{l} \text{ODE} \\ \text{SODE} \\ \text{RODE} \end{array} \right\} \Rightarrow \text{Sec 6.}$

Sec 2. - Existence of RODE.

3 Thm.

$$\frac{dx}{dt} = \underbrace{g(x, \eta_t)}_{(w)}$$

- 1) η_t : continuous $\Rightarrow$

- 2) η_t : measurable. $\Rightarrow$

1. Picard - Lindelöf

2. Peano's $x(t)$.

3. Carathéodory sense.

3. $x(t)$: Absol cont.

1.1. Simple numeric scheme for RODE.

$$\frac{dx}{dt} = -x + \eta_t.$$

- Euler scheme.

$$\int_{t_n}^{t_{n+1}} \Rightarrow x(t_{n+1}) - x(t_n) = \int_{t_n}^{t_{n+1}} (-x(s) + z_s) ds$$

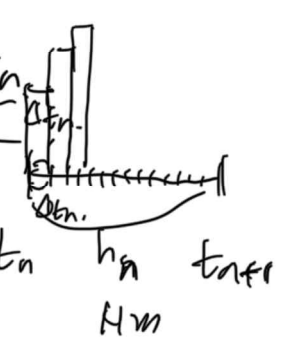
$$x(t_{n+1}) - x(t_n) = -x(t_n) + z_{t_n}$$

$$h_n := t_{n+1} - t_n$$

$$x_{n+1} = (1 - h_n) x_n + h_n z_{t_n}$$

- Averaged Euler scheme.

$$x_{n+1} = (1 - h_n) x_n + \sum_{j=1}^{N_{h_n}} z_{t_n + \frac{j-1}{N_{h_n}} h_n} \Delta t_n$$

$\int_{t_n}^{t_{n+1}} z_s ds$

 $t_n \quad h_n \quad t_{n+1}$
 N_{h_n}

- $\frac{dx}{dt} = f(x) + \theta(x) z_t$ - SODE

$$x_{n+1} = x_n + h_n f(x_n) + \theta(x_n) \int_{t_n}^{t_{n+1}} z_s ds$$

$$= x_n + (f(x_n) + \theta(x_n) I_n) h_n$$

$$\frac{dx}{dt} = q(x, z_t) \quad \frac{1}{h_n} \int_{t_n}^{t_{n+1}} z_s ds$$

$$x_{n+1} = x_n + \int_{t_n}^{t_{n+1}} q(x(s), z_s(w)) ds \quad \text{implicit.}$$

$$x_{n+1} = x_n + q(x_n, I_n) h_n$$

1.2 Taylor expansion:

- To derive higher order scheme
- Check the order of convergence of scheme.

1.2.1 ODE. $\frac{dx}{dt} = f(t, x) \rightarrow$

Total derivative $u(t, x(t))$

$$D u(t, x) := \frac{du}{dt}(t, x) + \underbrace{f(t, x)}_{\frac{dx}{dt} = f} \frac{dt}{dx}(t, x)$$

$$x^{(j)}(t) = \frac{d^j x}{dt^j} = D^{j-1} f(t, x(t))$$

$\frac{dx}{dt} = f(t, x)$ constant
 $x'(t) = f(t, x)$

where $D^0 = Id$.

Let f : $p+1$ times, conti-diff., $\exists T \in [t_0, t_0+h]$.

$$x(t_0+h) = x(t_0) + x'(t_0)h + \dots + \frac{x^{(p)}(t_0)}{p!} h^p + \frac{x^{(p+1)}(\tau)}{(p+1)!} h^{p+1}$$

Def). p -Taylor approximation of the solution value.

$$x(t_0+h) \approx x(t_0+h, x_0, h_0)$$

$$\underline{T}(t_0, x_0, h) = x_0 + \sum_{j=1}^p \frac{h^j}{j!} D^{j-1} f(t_0, x_0)$$

1.2.2. SODE case

Ω_t : Holder continuous. $x(t)$:

$$x(t) = x(t_0) + \int_{t_0}^t f(s, x(s)) ds$$

1st Taylor
p-th

$$= x_0 + \int_{t_0}^t f(t_0, x(t_0)) dt + \int_{t_0}^t \int_{t_0}^s Df(t, x(t)) dt ds$$

$\therefore x(t)$: does not need derivative

$$x(t) = x_0 + \sum_{j=1}^p D^{j-1} f(t_0, x_0) \int_{t_0}^t \int_{t_0}^{s_1} \dots \int_{t_0}^{s_{j-1}} ds_j \dots ds_1 \\ + \int_{t_0}^t \int_{t_0}^{s_1} \dots \int_{t_0}^{s_{j-1}} D^j f(s_j, x(s_j)) ds_{j+1} \dots ds_1$$

(2.3) RODE case z_t .

$$\frac{dx}{dt} = \underbrace{q(x, z_t(w))}_{\Omega}, \quad \underbrace{z_t = Gw(t, x)}_{\text{may not diff int.}}$$

q : cont fcn. w.r.t both variable

$t \rightarrow q(x(t), z_t(w(t)))$: not diff

$(x, w) \rightarrow \underline{q(x, w)}$: diff.

$$\underline{q(x(s), w(s))} = \sum_{|H| \leq p} \frac{1}{j!} \frac{\partial^j q(x_0, w_0)}{\partial x^1 \dots \partial x^j \partial w^1 \dots \partial w^j} (\underbrace{\Delta x_s}_{x(s) - x_0})^1 (\underbrace{\Delta w_s}_{w(s) - w_0})^2 + R_{p+1}(s)$$

$$\Delta x_t = x(t) - x_0 = \int_{t_0}^t \underline{q(x(s), w(s))} ds \\ = \int_{t_0}^t \sum_{|H| \leq p} \frac{1}{j!} \frac{\partial^j q(x_0, w_0)}{\partial x^1 \dots \partial x^j \partial w^1 \dots \partial w^j} (\Delta x_s)^1 (\Delta w_s)^2 ds$$

$$+ \int_{t_n}^t R_{p+1}(s) ds$$

$$\int_{t_n}^{t_{n+1}} \Rightarrow x_{n+1} - x_n = \int_{t_n}^{t_{n+1}} \sum_{|H| \leq p} \frac{1}{j!} \partial^j q(x_n, u_n) (\Delta x(s))^j (\Delta u(s))^j ds$$

When $p=0$.

$$x_{n+1} - x_n = h_n q(x_n, u_n)$$

When $p \geq 1$. Implicit

$p=2$
 $p=3$

When $p \geq 1$. $\Delta = [1,0) \quad \Delta = [0,0)$ ~~$\Delta = [0,1)$~~

$$x_{n+1} - x_n = \int_{t_n}^{t_{n+1}} \sum_{\substack{\Delta = [1,0) \\ \Delta = [0,0)}} \frac{1}{j!} q(x_n, u_n) (\Delta x(s))^j (\Delta u(s))^j ds$$

$$= h_n q(x_n, u_n) + \partial_x q(x_n, u_n) \int_{t_n}^{t_{n+1}} \Delta x(s) ds$$

$$\Delta x(s) = x(s) - x(t_n) = \int_{t_n}^s q(x(s), u(s)) ds$$

$$= (s - t_n) q(x_n, u_n)$$

$$= h_n q(x_n, u_n) + \frac{1}{2} h_n^2 \partial_x q(x_n, u_n) q(x_n, u_n)$$

Sec 2. Random ODE.

$$\frac{dx}{dt} = q(\underline{x}, \underline{u}_t(\omega)) := G_{\omega}(t, x)$$

$\gamma : [0, T] \times \Omega \rightarrow \mathbb{R}^n$; stochastic process

with const. bounded path

with some sample path.

$g: \mathbb{R}^d \times \mathbb{R}^m \rightarrow \mathbb{R}^d$; conti

- If g_t continuous in $t \rightarrow G$ cont t . $\Rightarrow$ Classic ODE Thm
- If g_t measurable in $t \rightarrow G$ meas t . $\Rightarrow$ Carathéodory sense

2.1.1. (classical assumption

Fix a sample path writ w .

$$\cdot \quad \frac{dx}{dt} = G(t, x) \quad x(t_0) = x_0 \quad (\text{diff eqn})$$

$$\cdot \quad \left(\underline{x(t)} = x_0 + \int_{t_0}^t \underline{G(s, x(s))} ds \right) \quad (\text{int eqn})$$

lem If $x(t)$; conti & satisfied (int eqn)

$\Rightarrow x(t)$; sol of (diff eqn) & $x(t)$; cont diff.

Thm 2.1, (Picard-Lindelöf)

Let $R = \{ (t, x) : t \in [t_0, t_0 + a], |x - x_0| \leq b \}$

G unif Lipschitz cont in x cont in t .

$|G(t, x)| \leq M$ on R .

$\Rightarrow$ (diff) eqn has unique sol x^* on $[t_0, t_0 + \mu]$

$\mu = \min \{ a, \frac{b}{M} \}$ (for fixed sample path)

Proof) (Step 1) define $x_n(t)$

$$\underline{x_{n+1}(t)} = x_0 + \int_{t_0}^t G(s, \underline{x_n(s)}) ds$$

with $x_0(t) = x_0$

$$\Rightarrow \begin{cases} x_{n+1}(t) : \text{cont on } [t_0, t_0 + \tau] (-: x_n : \text{cont}) \\ |x_{n+1}(t) - x_0| \leq b \end{cases}$$

$$\hookrightarrow \therefore |x_{n+1}(t) - x_0| \leq \underbrace{|t - t_0|} \cdot \underbrace{|G|}_M \leq M \tau \leq b$$

(Step 2)

$$|x_{n+1}(t) - x_n(t)| \leq \frac{M K^n (t - t_0)^{n+1}}{(n+1)!}$$

$x_n = x_{n-1}$

when $n \geq 0$

$$|x_1(t) - x_0(t)| \leq \int_{t_0}^t G(s, x_0(s)) ds \leq M (t - t_0)$$

when $n \geq 1$

$$|x_{n+1}(t) - x_n(t)| = \int_{t_0}^t \underbrace{G(s, x_n(s)) - G(s, x_{n-1}(s))}_{\text{Lipshitz}} ds$$

$$\leq K \int_{t_0}^t |x_n(s) - x_{n-1}(s)| ds \quad (\text{Lipshitz})$$

$$\leq K \int_{t_0}^t \frac{M K^{n-1} (s - t_0)^n}{n!} ds \quad (\text{Induction})$$

$$\leq M K^n \cdot \frac{(t - t_0)^{n+1}}{(n+1)!}$$

(step 3),

$$x_{n+1}(t) = x_0 + \sum_{k=0}^n (x_{k+1}(t) - x_k(t))$$

$$|x_m(t) - x_n(t)| \leq \sum_{i=n}^{m-1} |x_{i+1}(t) - x_i(t)| \leq \sum_{i=n}^{m-1} M e^{\frac{(t-t_0)^{i+1}}{(i+1)!}} < \varepsilon$$

$\exists N, \leq n, m \quad \{x_n\} : \text{unif Cauchy} \Rightarrow \text{unif}$

$$\lim_{n \rightarrow \infty} x_n(t) = x^*(t) \quad \text{unif on } [t_0, t_0 + \tau],$$

$$\lim_{n \rightarrow \infty} G(t, x_n(t)) = G(t, x^*(t)) \quad \text{unif}$$

$\therefore \{G\} \in M, \quad \text{DCT}$

$$x^*(t) = x_0 + \int_{t_0}^t G(s, x^*(s)) ds$$

(step 4) Uniqueness

Let $y(t)$ is another sol with $y(t_0) = x_0$

$$|x_n(t) - y(t)| \leq \frac{M e^{\eta(t-t_0)^{\eta+1}}}{(n+1)!}$$

$$n \rightarrow \infty, \quad x^*(t) = y(t)$$

Thm 2.2 (Peano's thm)

G : cont in x and t .

$|G(t, x)| \leq M$ on R

(continued from previous page)

$\Rightarrow$ (diff eqn) has at least one sol

x^* on $[t_0, t_0 + \mu]$ for $\mu = \min\{a, b/\mu\}$

Proof) $x_0(t)$, cont diff $[t_0 - \delta, t_0]$

- $x_0(t_0) = x_0$

- $|x_0(t) - x_0| \leq b$

- $|x_0(t) - x_0(s)| \leq M|t-s|$ for $t, s \in [t_0 - \delta, t_0]$

$x_\epsilon(t)$ for $0 < \epsilon \leq \delta$ on $[t_0 - \delta, t_0 + \mu]$

$$x_\epsilon(t) = \begin{cases} x_0(t) & t \in [t_0 - \delta, t_0] \\ x_0 + \int_{t_0}^t G(s, x_\epsilon(s-\epsilon)) ds & t \in [t_0, t_0 + \mu] \end{cases}$$

$\Downarrow$

continuous

$t \in [t_0, t_0 + \epsilon]$ $n\epsilon = t_-$

$\Rightarrow [t_0 + \epsilon, t_0 + 2\epsilon]$

(Step 2), estimate of $x_\epsilon(t)$

$$|x_\epsilon(t) - x_0| = \left| \int_{t_0}^t G(s, x_\epsilon(s-\epsilon)) ds \right| \leq M(t-t_0) \leq M\mu = b$$

$\Rightarrow \{x_\epsilon\}$: unif bdd.

$$|x_\epsilon(t) - x_\epsilon(s)| = \left| \int_s^t G(\tau, x_\epsilon(\tau-\epsilon)) d\tau \right| \leq M|t-s|$$

$\Rightarrow \{x_\epsilon\}$: equi cont.

There exist subsequence $\epsilon_1, \epsilon_2, \dots$ s.t. $\epsilon_n \rightarrow 0$ as $n \rightarrow \infty$. and

$$\lim_{n \rightarrow \infty} x_{\epsilon_n}(t) = x^*(t) \quad \text{unif on } [t_0 - \delta, t_0 + \epsilon_n]$$

$$\lim_{n \rightarrow \infty} G(t, x_{\epsilon_n}(t - \epsilon_n)) = G(t, x^*(t)) \quad //$$

Thus (DCT)

$$x^*(t) = x_0 + \int_{t_0}^t G(s, x^*(s)) ds$$

Chapter 2. Random ODE.

2.1.3. Carathéodory Assumption.

Brief review

Sec 1 Taylor expansion.

- ODE - deriv
- SODE - inte.
- RODE - $q(x, x)$,

Sec 2. RODE.

$$\frac{dx}{dt} = \underbrace{q(x, x)}_{\substack{\leftarrow \text{not} \\ \leftarrow \text{near}}} = G_{\text{near}}(t, x) \quad (\text{RODE})$$

where $\eta: [0, T] \times \Omega \rightarrow \mathbb{R}^m$ stochastic process,

$(\Omega, \mathcal{F}, \mathbb{P})$: probability space.

q, G : cont in both variable.

$$\|G\| \leq m.$$

Z_t : cont int.

- $\left\{ \begin{array}{ll} \bullet \text{ Picard-Lindelöf} & (G, \text{Lip in } x) \quad \text{uniqu} \\ \bullet \text{ Peano} & (G, \text{cont int}) \quad \text{not} \end{array} \right.$

Today

$$Z_t : \text{cont int.} \Rightarrow g, G_w : \text{cont int}$$

$$\Rightarrow x(t) \text{ sol (RDE)} \Rightarrow x : \text{cont diff.}$$

$$Z_t : \text{meas int.} \Rightarrow g, G_w : \text{mea in } t. \quad \text{absolutely cont}$$

$$\Rightarrow x(t) \text{ sol (RDE)} \Rightarrow x : \underbrace{(\cdot)}_{\text{absolutely cont}}$$

$$\int G_w(t, x) dt$$

$Z_t : \text{meas}$

Def) $x : [t_0, T] \rightarrow \mathbb{R}^d$: absolutely cont.

if for $\forall \epsilon > 0, \exists \delta > 0$ s.t.

$$\sum_{i=1}^n |x(t_i) - x(s_i)| < \epsilon \quad \text{for } \{[s_i, t_i) : i = 1, \dots, n\}$$

which is finite collection of non-overlapping intervals

$$\text{in } [t_0, T], \quad \underbrace{\sum_{i=1}^n (t_i - s_i)} < \delta.$$

Note x : absolutely cont $\Rightarrow x$: weak differentiable almost everywhere

weak deriv

$$\int_{t_0}^{t_0} x'(s) \phi(s) ds = - \int_{t_0}^T x(s) \phi'(s) ds$$

$$\phi \in C_0^\infty([t_0, T], \mathbb{R})$$

[Lem 2.3]. If $f : [t_0, T] \rightarrow \mathbb{R}$ bdd, Leb mea, Leb int.

$$\& \quad \underline{x}(t) = \int_{t_0}^t \boxed{f(s)} ds$$

$\Rightarrow x(t)$: absolutely cont. & it weak derivate $x'(t) = f(t)$,
for almost all $t \in [t_0, T]$.

(Last /, f : cont $\Rightarrow x(t) \Rightarrow$ cont diff)

If x : cont $t \rightarrow G(t, x(t))$ Leb int.

$$\underline{x(t) = x_0 + \int_{t_0}^t G(s, x(s)) ds} \quad \underline{(IVP)}$$

absol cont.

$$\frac{dx}{dt} = G(t, x(t)), \quad x(t_0) = x_0.$$

Def. A function G is said to satisfy Carathéodory Assup

(C1) (cont) $G(t, x)$: cont in x for almost every $t \in [t_0, T]$.

(C2) (meas) $G(t, x)$: Lebesgue mea. in t for each $x \in \mathbb{R}^d$

(C3) (bounded) $|G(t, x)| \leq m(t)$: absol cont. $t \in [t_0, T]$

Theorem (Carathéodory). Let G satisfy (C) conditions.

then the (IVP) has a solution, $x^* : [t_0, t_0 + \delta] \rightarrow \mathbb{R}^d$.

Proof). $\underline{M(t) = \int_{t_0}^t m(s) ds}$; $\begin{matrix} \leq t_0 \\ \oplus \end{matrix} \rightarrow t \in [t_0, T]$ cont, non-decreasing.
 $M(t_0) = 0.$

(Step 1) Construct $x_n(t)$. $t \in [t_0, t_0 + \delta]$.

For $\delta > 0.$

If $\delta = 0$ $\delta \rightarrow 0$

$\& \quad \underline{x_0}$. &

$t \in [t_0, t_0 + \frac{\delta}{n}]$

$$\underline{x_n(t)} := \begin{cases} x_0 + \int_{t_0}^{t - \frac{\delta}{n}} G(\omega, s, \underline{x_n(s, \omega)}) ds & t \in [t_0 + \frac{\delta}{n}, t_0 + \delta] \end{cases}$$

$$x_n(t) = x_0$$

$$t \in [t_0 + \frac{\delta}{n}, t_0 + 2 \cdot \frac{\delta}{n}] \\ \Rightarrow t \in [t_0 + \frac{2\delta}{n}, t_0 + 3 \cdot \frac{\delta}{n}]$$

$$x_2(t) = \begin{cases} x_0 & [t_0, t_0 + \frac{\delta}{2}] \\ x_0 + \int_{t_0}^{t - \frac{\delta}{2}} G(\omega, s, x_2(s, \omega)) ds & [t_0 + \frac{\delta}{2}, t_0 + \delta] \end{cases}$$

$x_n(t)$: cont function for $t \in [t_0, t_0 + \delta]$.

(step 2) Estimate of $x_n(t)$

By (3)

$$\bullet \quad \underline{|x_n(t) - x_0|} \leq \begin{cases} 0 & \text{if } t \in [t_0, t_0 + \frac{\delta}{n}] \\ \int_{t_0}^{t - \frac{\delta}{n}} G(s, x_n(s)) ds & \text{otherwise} \end{cases} \\ \leq \int_{t_0}^{t - \frac{\delta}{n}} M(s) ds \\ \leq \underline{M(t - \frac{\delta}{n})} \\ \leq \bar{M}.$$

$\therefore x_n(t)$: unif bdd

$$\bullet \quad \underline{x_n(t_1) - x_n(t_2)} = \begin{cases} 0 & t_1, t_2 \leq t_0 + \frac{\delta}{n} \\ \left(x_0 + \int_{t_0}^{t_1 - \frac{\delta}{n}} G ds \right) - x_0 & t_2 \leq t_0 + \frac{\delta}{n} \leq t_1 \\ \left(x_0 + \int_{t_0}^{t_1 - \frac{\delta}{n}} G ds \right) - \left(x_0 + \int_{t_0}^{t_2 - \frac{\delta}{n}} G ds \right) & t_0 + \frac{\delta}{n} \leq t_1, t_2 \end{cases}$$

$$\text{i) } \underline{|x_n(t_1) - x_n(t_2)|} \leq \left| M(t_1 - \frac{\delta}{n}) - M(t_2 - \frac{\delta}{n}) \right|$$

$$\text{ii) } t_1 - \frac{\delta}{n} \quad t_2 - \frac{\delta}{n} \quad t_1 > t_2$$

$$\int_{t_2 - \frac{\delta}{n}}^{t_1 - \frac{\delta}{n}} G \, ds \leq \int_{t_2 - \frac{\delta}{n}}^{t_1 - \frac{\delta}{n}} \underbrace{|G|}_{\leq m(t)} \, ds$$

$$= \left| M\left(t_1 - \frac{\delta}{n}\right) - M\left(t_2 - \frac{\delta}{n}\right) \right|,$$

$$ii) \int_{t_0}^{t_1 - \frac{\delta}{n}} G \, ds \leq \underbrace{M\left(t_1 - \frac{\delta}{n}\right)}_{t_2 - \frac{\delta}{n} \leq t_0} \left[- \underbrace{M\left(t_2 - \frac{\delta}{n}\right)}_{t_2 - \frac{\delta}{n} \leq t_0} \right]$$

$$|x_n(t_1) - x_n(t_2)| \leq M |t_1 - t_2|.$$

$\therefore \underline{x_n(t)}$ is equicontinuous.

(Step 3) Arzela-Ascoli

$$\lim_{n \rightarrow \infty} x_{n_f}(t) = x^*(t) \text{ unif on } [t_0, t_0 + \delta].$$

By (C1) G is cont in x for almost every fixed t .

$$\lim_{n \rightarrow \infty} G(t, x_{n_f}(t)) = G(t, x^*(t))$$

By (C3) $|G(t, x_{n_f}(t))| \leq m(t)$, By DCT,

$$\lim_{n \rightarrow \infty} \int_{t_0}^t G(s, x_{n_f}(s)) \, ds = \int_{t_0}^t G(s, x(s)) \, ds$$

$$\left(x_{n_f}(t) = x_0 + \int_{t_0}^t G(s, x_{n_f}(s)) \, ds - \underbrace{\int_{t_0}^{t - \frac{\delta}{n_f}} G(s, x_{n_f}(s)) \, ds}_{\substack{\Delta \\ 0, \quad |G| \leq m(t_2)}} \right)$$

+

$$\Rightarrow x^*(t) = x_0 + \int_{t_0}^t G(s, x^*(s)) ds$$

for all $t \in [t_0, t_0 + \delta]$.

Summary.

If x_t cont; G cont fn t $\Rightarrow \underline{x(t)}$ cont diff $\begin{matrix} \text{P-L} \\ \text{ZP} \end{matrix}$
 If x_t meas; G meas in t. $\Rightarrow x(t)$: absol cont
 $\Rightarrow$ Carathéodory